

DEEPAK PRASAD SHAH

Student • Full-Stack Developer • Innovator

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Available for internships & remote projects

PROFILE

BE Computer Science & Engineering student at Chandigarh University (batch 2028) with a passion for building systems that connect hardware, software and ideas. Experienced in full-stack web development, IoT and embedded systems, AI/ML, and product design. Holds 3 published patents in hardware and automation. Ranked 3rd out of 300 teams in a national-level hackathon. Comfortable taking ownership of projects end-to-end while also excelling in collaborative team environments.

EDUCATION

B.E. Computer Science & Engineering

2024 – 2028

Chandigarh University, Punjab, India

- CGPA: 7.5 / 10
- 3+ Published Patents in hardware, counter-UAV and renewable energy automation
- Top positions in campus-level hackathons; Recognised for high-impact engineering projects
- President, Chandigarh University Nepalese Student Society (CUNSS)

Senior Secondary – Grade 11 & 12 (PCM)

2022 – 2024

Kendriya Vidyalaya, Embassy of India (KV EOI), Kathmandu, Nepal

- Aggregate: 85% | Physics, Chemistry, Mathematics
- House Captain | Strong problem-solving foundation

Secondary – Class 1 to 10

up to 2022

Modern Indian School, Kathmandu, Nepal

WORK EXPERIENCE

Freelance Developer & Designer

2025 – Present

Self-employed | Remote

- Delivered 5+ custom portfolio websites for clients with unique themes, responsive layouts, animations and SEO-optimised content.
- Designed and launched a premium wedding invitation website with RSVP management.
- Developed functional web app for Happy Hills Restaurant (RMS) — solving core pain-points of existing solutions in the market.
- Built and delivered promotional websites for local businesses.
- Assisting in mypckey.com (WordPress-based); currently developing prudentialtrading.com.np and everlastmedcoregroup.com.

Tech: WordPress, React, Next.js, JavaScript, HTML/CSS, GSAP, SEO, Vite

Product Designer – CU Management System Staff Portal

2025

Chandigarh University | ERP Feature Design

- Designed and developed features for the staff-facing CU Management System 2.0 login portal (staff.cuchd.in).
- Focused on improving the technical support interface, UX, and ticket-raising functionality.

Tech: UI/UX Design, Figma, ERP Systems, Prototyping

PROJECTS

Agri_Doctor — AI-Powered Plant Health System

AI Developer

- 3-tier AI system detecting plant diseases, running interactive diagnostic Q&A (Akinator-style), and providing treatment plans with GEO-tagging for regional disease mapping.
- Self-trained ML models with OpenAI API fallback; real-time processing pipeline.

Tech: Python, FastAPI, Machine Learning, React, OpenAI API, GIS/GEO

Desert Offroad Navigator — Hackathon (Ranked 3 / 300)

AI/ML Developer

- Real-time terrain segmentation and NLP-based voice guidance for off-road drivers, integrating GPS and GEO-spatial hazard mapping.

Tech: Computer Vision, Segmentation, GPS/GIS, NLP, Python

Algolabs — EdTech Learning Platform

Full-Stack Developer

- Comprehensive platform for mastering Data Structures, Algorithms and JavaScript, with interactive algorithm visualisation tools animated in real time using the Canvas API.

Tech: React, D3.js, TypeScript, Node.js, Canvas API, SEO

CUNSS Election Portal — Secure Voting System

Lead Developer

- Online election portal for Chandigarh University Nepalese Student Society — Google OAuth-verified voting, real-time vote tallying, transparent result declaration.

Tech: React, Node.js, Google OAuth, Express.js, MongoDB

Screen Sharing Tool — Real-time Web App

Lead Developer

- Browser-based screen sharing enabling teachers to create secure rooms and students to join via room code; WebRTC for low-latency P2P streaming, deployed on Render with Docker.

Tech: JavaScript, WebRTC, Socket.IO, Node.js, Express.js, Docker, Render

Restaurant Management Portal (RMS)

Lead Developer

- Full-stack portal handling table orders, menu management and kitchen workflows with separate interfaces for customers, waiters and kitchen staff.

Tech: Flask, Python, HTML/CSS, MySQL, REST APIs

IoT-Based Smart Irrigation System

Lead Developer

- End-to-end smart irrigation using ESP32, soil moisture sensing, temperature data and rainfall prediction to automate watering and cut water consumption.

Tech: ESP32, IoT, C/C++, Sensors, REST API

Scam-Guard Browser Extension

Lead Developer

- Web extension for real-time HTML scraping and analysis to detect and alert users of phishing attempts and online scams.

Tech: JavaScript, Browser Extension API, HTML Scraping, Security Analysis

Meal Planner App

Full-Stack Developer

- Web app for planning weekly meals, tracking ingredients and minimising food waste; integrated with WordPress for dynamic nutritional content.

Tech: React, Node.js, MongoDB, WordPress

Examination Portal

Developer

- Online portal managing question banks, student authentication, timed tests and automatic result calculation.

Tech: Python, Flask/Django, MySQL, Authentication

Student Management System

Developer

- Desktop/web application for managing student data, course enrolments and basic analytics with focus on performance and clean data structures.

Tech: Python, Flask, SQL, Data Structures

PUBLISHED PATENTS & INNOVATIONS

Drone Immobilisation Over 2.4GHz

Flagship Patent

Counter-UAV | Wireless Jamming | Security | RF Engineering

- Design and implementation of a system that safely immobilises unauthorised drones on 2.4GHz bands via controlled disruption without collateral interference.

Solar Panel Surface Cleaning System

Published Patent

Renewable Energy | Automation

- Automated cleaning mechanism for solar panels that maximises power output by reducing dust accumulation with minimal water usage and manual intervention.

Mechanised Lifting System

Published Patent

Mechanical Design | Safety

- Ergonomic mechanised lifting solution for safely handling loads in constrained spaces, focused on safety and ease of operation.

SKILLS

Programming & Software

- Python
- C / C++
- JavaScript
- React
- Next.js
- Node.js
- Flask / Django
- WordPress
- Git & GitHub

Embedded & IoT

- ESP32 / Arduino
- Sensors & Actuators
- Wireless Communication (2.4GHz)
- IoT Protocols
- REST APIs for IoT

Systems & Concepts

- Data Structures & Algorithms
- System Design
- Product Design

Design & Other

- UI/UX Design (Figma)
- Technical Writing
- Patent Drafting

- RESTful APIs
- WebRTC / Socket.IO
- Docker
- Public Speaking
- Team Collaboration
- SEO Optimisation

LEADERSHIP & ACTIVITIES

President – CUNSS (Chandigarh University Nepalese Student Society) 2024 – 2026

- Led the Nepalese student community at Chandigarh University; organised and managed events attended by 450+ students.
- Anchored multiple community and cultural events, representing the Nepalese student body on campus.
- Won several university-level Group Discussions (GDs), demonstrating strong communication and critical thinking.
- Built the CUNSS Election Portal for secure, transparent community governance.

House Captain 2022 – 2024

Kendriya Vidyalaya, Embassy of India, Kathmandu

- Represented the house in academic and co-curricular activities; led team coordination and event planning.